

EDUCATION

- National Institute of Technology Jalandhar** Punjab, India
• *Bachelor of Technology (Honours) - Electronics and Communication; CGPA: 8.53* 2022 - 2026
Courses: Information Theory, Error Control Coding, Digital Communication Systems, Antennas and Wave Propagation, Analog and Digital Communication Systems

EXPERIENCE

- Indian Institute of Technology Madras — TelWise Research Group** Chennai, On-site
• *Project Associate (Under Prof. K. Giridhar)* Jul 2026 - Present
 - **RFSoc-Based HF Radio Transceiver Implementation:** Implementing transmit and receive signal chains for WALE HF radios on the AMD/Xilinx Zynq UltraScale+ RFSoc ZCU216, including digital baseband processing and RF data-converter interfacing.
- Indian Institute of Technology Madras — TelWise Research Group** Chennai, On-site
• *Research Intern (Under Prof. K. Giridhar)* Jan 2026 – Jun 2026
 - Completed B.Tech thesis on “Non-Minimum-Phase Channel DoA Estimation via UCA Phase-Mode Smoothing and Enhanced Subspace Separation”.
- Indian Institute of Technology Madras — TelWise Research Group** Chennai, On-site
• *Summer Research Fellow (Under Prof. K. Giridhar)* Jun 2025 – Aug 2025
 - **Advanced DoA Estimation for Coherent Sources using UCAs:** Worked on implementing and benchmarking eigenstructure based array signal processing methods for passive radar applications for Uniform Circular Arrays (UCAs).
- Corps of Signals, Indian Army** Jalandhar, On-site
• *Project Intern (Part-time, Under Dr. Sukwinder Singh)* Aug 2024 – Oct 2024
 - **GNSS SDR Front-End Configuration and Acquisition Validation:** Configured an RTL-SDR + GPS patch antenna and GNU Radio acquisition chain to capture raw GNSS baseband data. Used GNSS-SDR on Linux to extract CAF plots for detected satellites and validate the lab’s in-house acquisition module, confirming PRN detection.

PUBLICATIONS

- Non-Minimum-Phase Channel DoA Estimation and Enhanced Subspace Separation** Oct 2026
• *Conference: IEEE MILCOM 2026* Accepted
- RDSE: Real-Time Directional Speech Enhancement Framework for Dual-Mic Smartphones** Oct 2026
• *Conference: IEEE IECON 2026* Accepted
- Real-Time Non-Contact Soil Moisture Estimation Using Machine Learning at the Edge** Dec 2025
• *Conference: IEEE ICEdge 2025, IISc Bangalore* [link](#)
- Comparative Analysis of U-Net and U-Net++ for Solar Potential Estimation** Nov 2025
• *Conference: IEEE INSPECT 2025 (Best Paper – Signal and Image Processing Track)* [link](#)

PROJECTS

- RDSE: Real-Time Directional Speech Enhancement Framework for Dual-Mic Commodity Smartphones:**
[Presentation and Live Demo](#)
 - Designed an audio-visual zooming framework combining traditional signal processing and modern machine learning for commodity dual-microphone smartphones.
- Warehouse Automation Bot Controller:**
[Github Link](#)
 - Designed and implemented a proof-of-concept autonomous warehouse bot controller using Verilog and MIMO digital design principles.

AWARDS AND ACHIEVEMENTS

- **Highest B.Tech Thesis Grade:** Awarded over a cumulative of three evaluations.
- **MILCOM Conference Travel Grant:** Awarded a grant worth \$1975 for presenting my work at MILCOM 2026.
- **Samsung ISWDP Fellowship:** Received a 100% waiver for the first, second and advanced levels of the ninth ISWDP cohort.
- **Best Paper Award:** Received the Best Paper Award in the Signal and Image Processing Track at IEEE INSPECT 2025
- **Selected for Direct Ph.D. admission (Signal Processing and Communications Group), IIT Kanpur — 2025:**